📊 Financial Performance Analysis using Python & Microsoft Power BI 📊

📝 Introduction:

Understanding financial performance is crucial for businesses to make informed decisions. This project focuses on analyzing sales, profit, and overall business performance using financial data. By leveraging Python for data cleaning and Microsoft Power BI for visualization, this project provides meaningful insights into revenue trends, profitability, and business segments.

🎯 Project Overview:

The Financial Performance Analysis project aims to evaluate business performance by analyzing key financial metrics such as revenue, profit, and profit margin. The dashboard helps stakeholders understand business trends, identify high-performing areas, and make data-driven decisions.

❗ Problem Statement:

Businesses often struggle to:

- Track revenue and profitability effectively
- Identify top-performing products and regions
- Understand the impact of discounts on profit
- Analyze monthly and segment-wise performance

This project solves these problems by creating an interactive dashboard that highlights:

- Revenue trends
- Profitability analysis
- Segment performance
- Product-level insights

📂 Dataset Information:

The dataset contains financial transaction data with key attributes:

📌 Data Definition:

- **Segment:** Customer segment (Government, Midmarket, etc.)
- **Country:** Country where sales occurred
- **Product:** Product name
- **Discount Band:** Discount category
- **Units Sold:** Number of units sold
- **Manufacturing Price:** Cost of production
- **Sale Price:** Selling price per unit
- **Gross Sales:** Total sales before discount
- **Discounts:** Discount amount

- **Sales:** Net sales after discount
- **COGS:** Cost of goods sold
- **Profit:** Net profit
- **Date:** Transaction date
- **Month Number:** Month (1–12)
- **Month Name:** Month name
- **Year:** Year

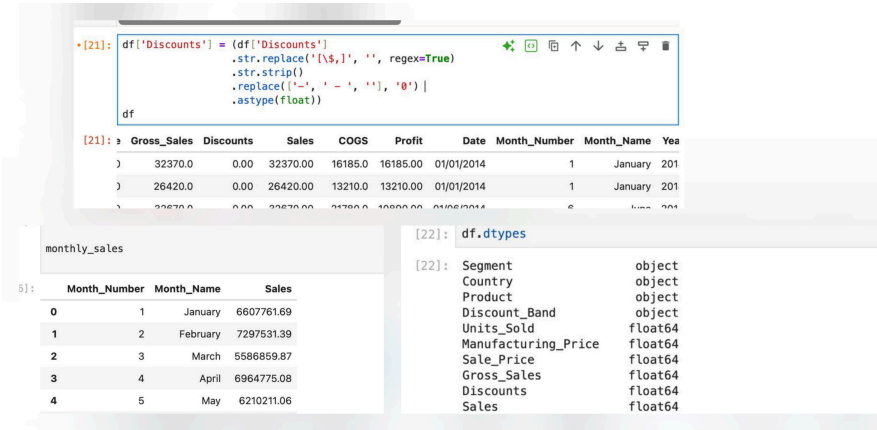
🔄 Project Workflow:

1️⃣ Data Collection:

- Imported dataset into Python environment

2️⃣ Data Preprocessing:

- Removed symbols like $ and ,
- Converted columns into numeric format
- Handled missing values
- Fixed data types
- Created new columns:
- Profit Margin
- Revenue per Unit



```
[21]: df['Discounts'] = (df['Discounts']
                         .str.replace('[\$,]', '', regex=True)
                         .str.strip()
                         .replace(['–', ' – ', ''], '0') |
                         .astype(float))
      df
```

[21]:	Gross_Sales	Discounts	Sales	COGS	Profit	Date	Month_Number	Month_Name	Yea
)	32370.0	0.00	32370.00	16185.0	16185.00	01/01/2014	1	January	201
)	26420.0	0.00	26420.00	13210.0	13210.00	01/01/2014	1	January	201

```
monthly_sales
```

5]:	Month_Number	Month_Name	Sales
0	1	January	6607761.69
1	2	February	7297531.39
2	3	March	5586859.87
3	4	April	6964775.08
4	5	May	6210211.06

```
[22]: df.dtypes
```

[22]:	
Segment	object
Country	object
Product	object
Discount_Band	object
Units_Sold	float64
Manufacturing_Price	float64
Sale_Price	float64
Gross_Sales	float64
Discounts	float64
Sales	float64

3️⃣ Exploratory Data Analysis (EDA):

- Analyzed sales by country
- Evaluated product-wise profit
- Studied monthly sales trends
- Identified segment performance

Segment	Country	Product	Discount_Band	Units_Sold	Manufacturing_Price	Sale_Price	Gross_Sales	Discounts	Sales	COGS	Profit	Date	Month_Number	Month_Name	Year	Profit_Margin
Government	Canada	Carretera	None	1618.5	3.0	20.0	32370.0	0.0	32370.0	16185.0	16185.0	01/01/2014	1	January	2014	0.5
Enterprise	Germany	Amarilla	High	3165.0	260.0	125.0	395625.0	43518.75	352106.25	379800.0		01/01/2014	1	January	2014	
Government	Mexico	Amarilla	High	2629.0	260.0	20.0	52580.0	5783.8	46796.2	26290.0	20506.2	01/01/2014	1	January	2014	0.43820224719101125
Small Business	United States of America	Paseo	High	3495.0	10.0	300.0	1048500.0	125820.0	922680.0	873750.0	48930.0	01/01/2014	1	January	2014	0.05303030303030303
Government	France	Carretera	High	2521.5	3.0	20.0	50430.0	6051.6	44378.4	25215.0	19163.4	01/01/2014	1	January	2014	0.4318181818181818
Government	United States of America	Montana	High	982.5	5.0	350.0	343875.0	44703.75	299171.25	255450.0	43721.25	01/01/2014	1	January	2014	0.14614121510673234
Small Business	Germany	Paseo	High	807.0	10.0	300.0	242100.0	31473.0	210627.0	201750.0	8877.0	01/01/2014	1	January	2014	0.0421455938697318
Enterprise	Mexico	VTT	High	554.0	250.0	125.0	69250.0	7617.5	61632.5	66480.0		01/01/2014	1	January	2014	
Midmarket	France	Velo	High	3997.5	120.0	15.0	59962.5	7795.13	52167.38	39975.0	12192.38	01/01/2014	1	January	2014	0.2337165485404864
Government	Canada	Montana	High	2227.5	5.0	350.0	779625.0	109147.5	670477.5	579150.0	91327.5	01/01/2014	1	January	2014	0.1362126245847176
Small Business	France	Paseo	Low	2434.5	10.0	300.0	730350.0	21910.5	708439.5	608625.0	99814.5	01/01/2014	1	January	2014	0.140893470790378
Government	France	Paseo	Low	3945.0	10.0	7.0	27615.0	276.15	27338.85	19725.0	7613.85	01/01/2014	1	January	2014	0.2784992784992785
Channel Partners	Canada	VTT	Medium	3244.5	250.0	12.0	38934.0	2725.38	36208.62	9733.5	26475.12	01/01/2014	1	January	2014	0.7311827956989246
Government	Mexico	Velo	None	1493.0	120.0	7.0	10451.0	0.0	10451.0	7465.0	2986.0	01/01/2014	1	January	2014	0.2857142857142857
Government	Germany	Carretera	None	1321.0	3.0	20.0	26420.0	0.0	26420.0	13210.0	13210.0	01/01/2014	1	January	2014	0.5
Enterprise	Canada	Amarilla	High	1659.0	260.0	125.0	207375.0	26958.75	180416.25	199080.0		01/01/2014	1	January	2014	
Channel Partners	France	VTT	High	1734.0	250.0	12.0	20808.0	2288.88	18519.12	5202.0	13317.12	01/01/2014	1	January	2014	0.7191011235955057
Government	United States of America	Paseo	High	1438.5	10.0	7.0	10069.5	1309.04	8760.47	7192.5	1567.97	01/01/2014	1	January	2014	0.17898240619510142
Government	Germany	Montana	High	766.0	5.0	350.0	268100.0	29491.0	238609.0	199160.0	39449.0	01/01/2014	1	January	2014	0.1653290529695024
Small Business	Canada	Paseo	High	873.0	10.0	300.0	261900.0	28809.0	233091.0	218250.0	14841.0	01/01/2014	1	January	2014	0.06367041198501873
Government	Canada	Paseo	Low	4251.0	10.0	7.0	29757.0	1190.28	28566.72	21255.0	7311.72	01/01/2014	1	January	2014	0.25595238095238093
Government	Germany	Paseo	Medium	1372.0	10.0	7.0	9604.0	480.2	9123.8	6860.0	2263.8	01/01/2014	1	January	2014	0.24812030075187974
Government	France	Montana	Medium	1384.5	5.0	350.0	484575.0	24228.75	460346.25	359970.0	100376.25	01/01/2014	1	January	2014	0.21804511278195488
Midmarket	United States of America	Velo	Medium	555.0	120.0	15.0	8325.0	416.25	7908.75	5550.0	2358.75	01/01/2014	1	January	2014	0.2982456140350877
Midmarket	Mexico	Velo	Medium	2861.0	120.0	15.0	42915.0	2145.75	40769.25	28610.0	12159.25	01/01/2014	1	January	2014	0.2982456140350877
Government	United States of America	Carretera	Medium	1117.5	3.0	20.0	22350.0	1341.0	21009.0	11175.0	9834.0	01/01/2014	1	January	2014	0.46808510638297873
Enterprise	France	Amarilla	Medium	1987.5	260.0	125.0	248437.5	14906.25	233531.25	238500.0		01/01/2014	1	January	2014	
Channel Partners	Germany	VTT	Low	2479.0	250.0	12.0	29748.0	892.44	28855.56	7437.0	21418.56	01/01/2014	1	January	2014	0.7422680412371134
Small Business	Mexico	Paseo	Medium	2565.0	10.0	300.0	769500.0	69255.0	700245.0	641250.0	58995.0	01/01/2014	1	January	2014	0.08424908424908426
Government	Mexico	Paseo	Medium	2417.0	10.0	350.0	845950.0	76135.5	769814.5	628420.0	141394.5	01/01/2014	1	January	2014	0.1836734693877551
Channel Partners	United States of America	VTT	Medium	1956.0	250.0	12.0	23472.0	2112.48	21359.52	5868.0	15491.52	01/01/2014	1	January	2014	0.7252747252747253

4️⃣ Dashboard Development:

- Imported cleaned data into Microsoft Power BI

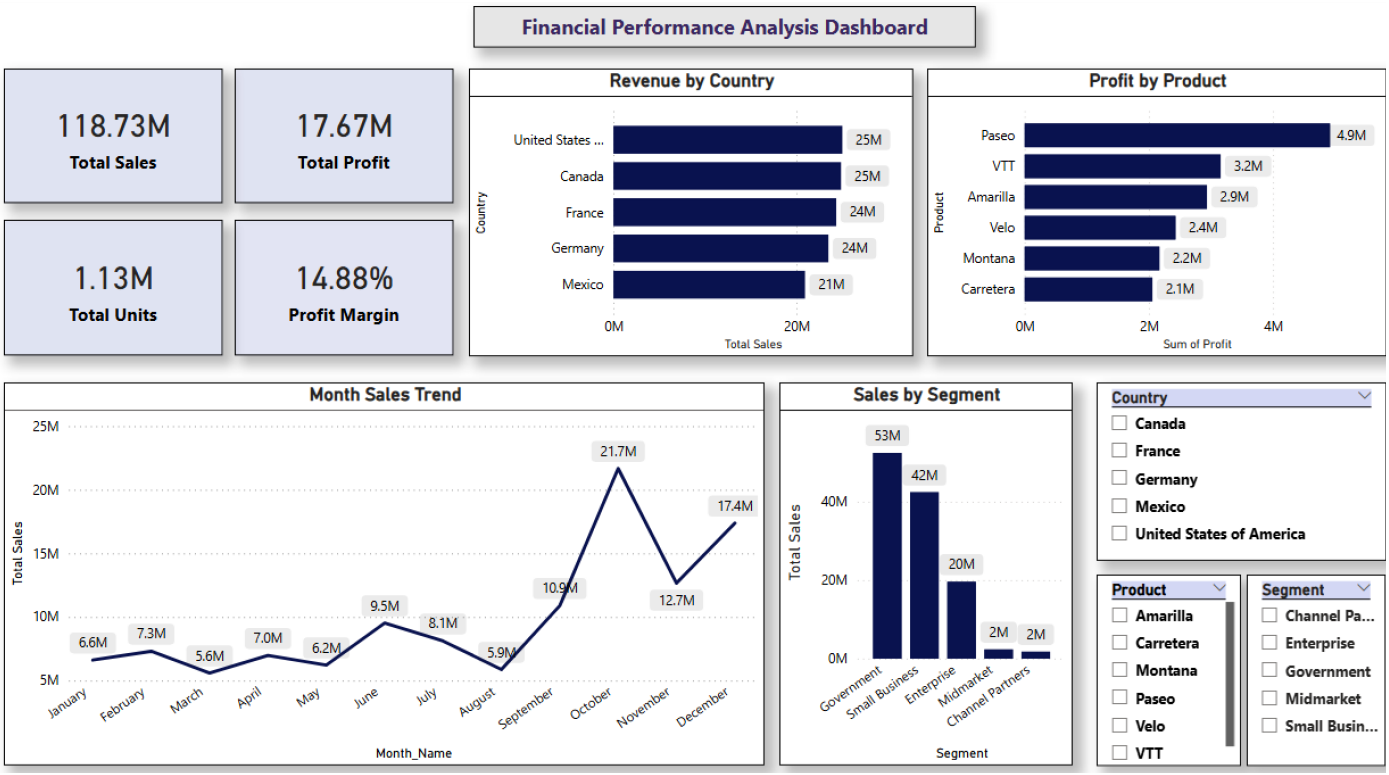
Created KPIs:

- Total Sales
- Total Profit
- Total Units Sold
- Profit Margin

Built interactive visuals:

- Sales by Country
- Profit by Product
- Monthly Sales Trend
- Sales by Segment

Added slicers for filtering



5️⃣ Business Insights:

- Identified top-performing countries and products
- Analyzed seasonal sales trends
- Evaluated impact of discounts on profitability
- Highlighted high-revenue segments

🛠️ Dependencies:

Ensure the following libraries are installed:
pip install pandas numpy

📚 Libraries Used:

- Python 3.x
- Pandas
- NumPy
- Microsoft Power BI

📑 Table of Contents:

1. Import Libraries
2. Load Dataset
3. Data Cleaning
4. Data Transformation
5. Financial Metrics Creation
6. Exploratory Data Analysis
7. Dashboard Development

📊 Key Features:

✔ Data cleaning using Python
✔ Financial KPI calculation
✔ Interactive dashboard
✔ Business performance insights
✔ Dynamic filtering using slicers

🚀 Usage Instructions:

1. Download the dataset
2. Run Python script / Jupyter Notebook for data cleaning
3. Import cleaned dataset into Power BI
4. Build dashboard visuals

📈 Results:

- Successfully analyzed financial data
- Identified revenue and profit trends
- Built an interactive dashboard for decision-making
- Derived actionable business insights

🔮 Future Enhancements:

- Add forecasting (time-series analysis)
- Implement advanced DAX calculations
- Include scenario analysis (budget vs actual)
- Deploy dashboard online

📌 Conclusion:

This project demonstrates how financial data can be transformed into meaningful insights using Python and Power BI. It highlights the importance of data cleaning, KPI creation, and visualization in financial analysis.

The dashboard helps businesses:

- Track performance
- Identify growth opportunities
- Make informed decisions

📧 Feedback and Suggestions:

Thank you for visiting this project! 😊
I would love to hear your feedback and suggestions.

✨ Support this project!

If you found this project useful:
⭐ Give it a star on GitHub.
📢 Share it with others.